

Exam in Statistical Physics and Solid State Physics

Exam date: 13.1.2023

Exam duration: 4 hours.

Permitted help: books, notes, computer without internet connection.

Answers can be given in English or Danish.

1 Lattice (3 points)

Figure 1 shows two two-dimensional lattices.

1. By drawing the smallest possible unit cell into the each lattice, show that the left lattice has one atom per unit cell and the right has two atoms per unit cell. (2 p)
2. Note that the right lattice is actually obtained from the left one by removing some of the atoms. How much larger (or smaller) is the unit cell on the right hand side with respect to the unit cell on the left hand side? (1 p)

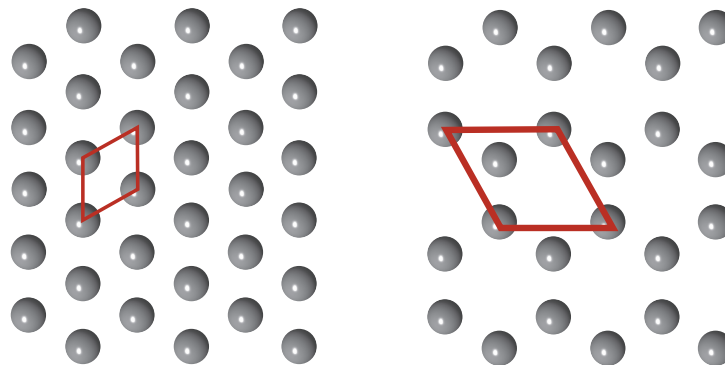


Figure 1: Two two-dimensional lattices.

The right unit cell is three times bigger than the left unit cell. This is easy to see by moving the unit cell from the right lattice onto the left lattice.

2 X-ray diffraction (2 point)

Figure 2 shows the reciprocal lattice of a crystal. Consider an X-ray diffraction experiment on this crystal.

1. Draw a wave vector for the incoming X-rays and a wave vector for the scattered X-rays into the figure such that the condition for constructive interference is fulfilled. (1.5 p.)
2. For the situation you have sketched in the figure, in which direction would the detector need to be placed in order to observe constructive interference? (0.5 p)

It would need to be placed in the direction of k_{out} .

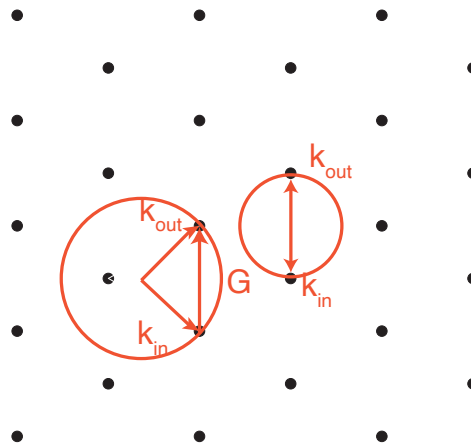


Figure 2: Reciprocal lattice of a crystal. Two solutions. Note that this is a drawing in reciprocal space. You can't draw wave fronts on top like in a Bragg picture.

3 Periodic boundary conditions (1 point)

Consider a one-dimensional solid with 100 unit cells and a lattice constant of 0.5 nm. The use of periodic boundary conditions restricts the possible k values and thus the allowed wave lengths for vibrations. Is a lattice wave with a wave length of 3 nm allowed given the periodic boundary conditions or not? Explain why.

The allowed wave lengths are $\lambda = L/n$ where L is the length of the chain and n is an integer. This follows from the allowed k values in equation (4.21) in the book. Since $L=50$ nm, 3 nm is not an allowed wave length.

4 Vibrations (2 points)

Consider a one-dimensional chain of atoms with one atom per unit cell and a lattice constant a .

1. For a very small wave vector of $k \ll 2\pi/a$, describe the relative motion of an arbitrary atom in the chain and an atom removed by 6 unit cells.
2. For a wave vector of $k = \pi/a$, describe the relative motion of an arbitrary atom in the chain and an atom removed by 6 unit cells.

In both cases, the arbitrary atom and the atom 6 unit cells away vibrate in phase.

5 Interatomic potential (1 point)

Consider two atoms separated by a distance R . Assume that the equilibrium distance is a . Give an expression for an interatomic potential $U(R, a)$ that is purely harmonic.

E.g. $U(R) = \alpha(R - a)^2 - \beta$.

6 Heat capacity (2 points)

1. Which element has the higher Debye temperature, Si or Pb? Give a qualitative explanation of why this is so. (1 p)
2. Taking your answer in the first part as a starting point, which element has the higher molar heat capacity at $T=50$ K and why? (1 p)

Si has the higher Debye temperature because lighter atoms and stronger forces. The Debye temperatures for these two elements are actually given in the book. Pb has the higher heat capacity at 50 K because of the lower Debye temperature.

7 Electronic states in crystals (1 point)

Consider a Bloch wave function $\psi_k(x)$ of an electronic state in a one-dimensional crystal with lattice constant a for a wave vector $k = \pi/a$. Assume that this wave function has the value α for $x = 0$. What is its value for $x = a$?

$-\alpha$. Note that the Bloch wave function does *not* have the periodicity of the lattice!

8 Band structure (2 points)

Figure 3 shows the band structure of a semiconductor along different directions in the Brillouin zone. The location of the band gap is marked in light grey.

1. How many valence electrons are there at least per unit cell? (1 p)
2. When referring to the bands near the valence band maximum, semiconductor physicists often speak of the “light hole band” and the “heavy hole band”. Mark in the figure which is which. (1 p)

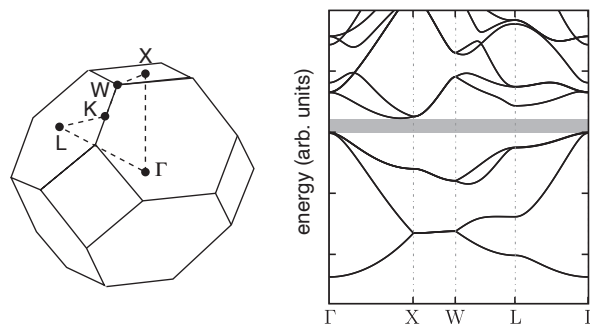


Figure 3: Band structure of a semiconductor.

The material contains at least 8 valence electrons per unit cell because there are at least 4 filled bands. The heavy hole band is the one with a flat parabola and the light hole band is the one with a steep parabola around Gamma in the valence band.

9 Superconductivity (2 points)

Lead has four stable isotopes with the following abundance: ^{204}Pb (1.4%), ^{206}Pb (24.1%), ^{207}Pb (22.1%) and ^{208}Pb (52.4%).

1. Suppose that you could enrich the heaviest isotope to 99% content. Would the resulting sample have a higher or a lower superconducting T_c compared to a sample of the natural isotope composition? Give a short reason for your answer.
2. Estimate if T_c would change by more than 1 K or not. T_c for its natural isotopic composition is 7.2 K.

Lower because the higher mass (on average) reduces the vibrational frequencies and thereby also T_c . The average mass of the natural isotopic composition is 207.241. The mass of the nearly isotope pure case is 208 (ignore the last %). So the ratio between the two is $207.24 / 208$ and the temperature changes by the square root of that times 7.2 K. That's less than 0.02 K. Or more simply, the worst-case ratio is $204 / 208$. Even if I don't take the square root of this, the change would only be 2% and so it would be much smaller than 1 K.

Problem 10

1) $N(N+1)/2$

2) $A = \epsilon, B = 1, C = 2\epsilon$

3) $Z \approx \frac{(kT)^2}{2\epsilon^2} = \frac{1}{2}Z_1^2$. For $kT \gg \epsilon$, there are many single-particle states available compared to the number of fermions, and hence the chance that the two fermions try to occupy the same single-particle state is negligible. The partition function is then simply the product of the partition functions for the individual particles, i.e. Z_1 times Z_1 , except that we need to divide by two, since we obtain the same system state if we exchange the fermions.

Problem 11

1)

$$T = \frac{T_1 C_{V,1} + T_2 C_{V,2}}{C_{V,1} + C_{V,2}}$$

2)

$$\Delta S = \int_{T_1}^T \frac{C_{V,1}}{T} dT + \int_{T_2}^T \frac{C_{V,2}}{T} dT = C_{V,1} \ln(T/T_1) + C_{V,2} \ln(T/T_2)$$

Problem 12

1)

$$\frac{e^{-\beta(\epsilon_A - \mu_A)} + 2e^{-\beta(\epsilon_{2A} - 2\mu_A)}}{1 + e^{-\beta(\epsilon_A - \mu_A)} + e^{-\beta(\epsilon_{2A} - 2\mu_A)} + e^{-\beta(\epsilon_B - \mu_B)}}, \quad \beta = \frac{1}{kT}$$

$$\mu_A = -kT \ln \left[\frac{1}{n_A} \left(\frac{2\pi m_A kT}{h^2} \right)^{3/2} \right], \quad \mu_B = -kT \ln \left[\frac{2}{n_B} \left(\frac{2\pi m_B kT}{h^2} \right)^{3/2} \right],$$

where k is Boltzmann's constant and h is Planck's constant.